# Preliminary Program

8th International Induced Polarization Workshop
May 26–28, 2026
Saint-Alexis-des-Monts, Québec, Canada
**https://internationalipworkshop.com/**

## Workshop schedule

**Tuesday, May 26, 2026**

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| 08:00–08:45 | Participant welcome |
| 08:45–09:00 | Welcome address |
| 09:00–10:30 | Fundamentals of IP |
| 10:30–11:00 | Coffee break |
| 11:00–12:30 | Emerging IP applications |
| 12:30–14:00 | Lunch |
| 14:00–15:30 | Outdoor IP networking and site exploration |
| 15:30–16:00 | Coffee break |
| 16:00–17:30 | Inversion of IP data at field scale |
| 19:00 | Group dinner at the Auberge |

**Wednesday, May 27, 2026**

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| 09:00–10:30 | Reactive processes controlling IP |
| 10:30–11:00 | Coffee break |
| 11:00–12:30 | Machine learning and IP data processing |
| 12:30–14:00 | Lunch |
| 14:00–15:30 | Dedicated poster session |
| 15:30–16:00 | Coffee break |
| 16:00–17:30 | Mineral exploration and groundwater studies |
| 19:00 | Group dinner at the Auberge |
| 21:00 | Bonfire (& Marshmallow) Networking Event |

**Thursday, May 28, 2026**

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| 09:00–10:30 | IP modelling and upscaling |
| 10:30–11:00 | Coffee break |
| 11:00–12:30 | Material and interfacial controls on IP |
| 12:30–14:00 | Lunch |
| 14:00–15:30 | Dedicated poster session |
| 15:30–16:00 | Coffee break |
| 16:00 | Closing remarks and workshop end |

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## Oral presentation sessions

Talks are 20 minutes long, including time for questions. Each session is 90 minutes.

### Fundamentals of IP

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| Surface Electrical Conductivity of Rocks and Soils: Novel Appraisal Ratios Beyond the Dukhin Number | Youzheng Qi<br>*Lawrence Berkeley National Laboratory* |
| The constrictivity exponent as a useful parameter in permeability prediction | Andreas Weller<br>*Clausthal University of Technology* |
| Estimating the Volume Fraction of Polarizable Material using IP chargeability | James Macnae<br>*RMIT University and CD3D* |
| Spectral induced polarization of sandstones | Konstantin Titov<br>*St. Petersburg University* |

### Emerging IP applications

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| Linking foam electrical conductivity to its electrochemical evolution during aging: a mechanistic approach | Pauline Kessouri<br>*BRGM* |
| Laboratory experiments for the investigation of low-frequency polarization effects at ice–water interfaces | Clemens Moser<br>*TU Wien* |
| Spectral Induced Polarization for Monitoring Geochemical Reactions during Subsurface Remediation | Hilary Emerson<br>*Pacific Northwest National Laboratory* |
| Can sulfide heap bioleaching be monitored by spectral induced polarization? Insights from laboratory column experiments | Philippe Leroy<br>*BRGM* |

### Inversion of IP data at field scale

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| Large-scale Induced Polarization Inversion of Mineral Residues Structures using High-Order Regularization on Unstructured Finite Element Meshes | Yan Zhou<br>*The University of Queensland* |
| 3-D EM inversion considering induced polarization effect | Gianluca Fiandaca<br>*University of Milano* |
| On the Inversion of Temperature Fields from IP Survey Data using unstructured FEM meshes | Lutz Gross<br>*The University of Queensland* |
| Joint inversion of inductive and galvanic data: significant resolution improvement honouring both data types – 3D EM and 2D DCIP | Gianluca Fiandaca<br>*University of Milano* |

### Reactive processes controlling IP

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| Mechanistic Insights into Organic Adsorption on Activated Carbon Using Spectral Induced Polarization | Nimrod Schwartz<br>*The Hebrew University of Jerusalem* |
| Using Microfluidics to Identify Mechanisms Controlling Induced Polarization Signatures during Hydroxyapatite Precipitation | Klaudio Peshtani<br>*Pacific Northwest National Laboratory* |
| Calcite mineralization dynamics with micromodel visualization and SIP monitoring | Sheng Zhou<br>*Zhejiang University* |
| On the fate and transport of heavy metals in calcareous soils – a spectral induced polarization study | Alex Furman<br>*Israel Institute of Technology* |

### Machine learning and IP data processing

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| Debye decomposition as a machine learning problem | Charles Bérubé<br>*Polytechnique Montréal* |
| Rapid Coal-Ash Characterization using Geophysical Methods & Machine Learning | Klaudio Peshtani<br>*Pacific Northwest National Laboratory*<br>*Rutger's University Newark* |
| A comparison between recurrent and feedforward neural networks for full-waveform TDIP denoising | Isabelle Poliquin<br>*Polytechnique Montréal* |
| Triple-Zarc Analysis of Electrical Impedance Spectra | Randy Enkin<br>*Geological Survey of Canada* |

### IP for mineral exploration and groundwater studies

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| Nodal-Based 3D Induced Polarization: A Comprehensive Workflow for Advanced Mineral Deposit Investigation | Bitnarae Kim<br>*BRGM* |
| The Role of 3D DCIP in the Leviathan Porphyry Discovery and Exploration | Jonathan Rudd<br>*Dias Geophysical* |

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| Linking Time-Lapse DCIP to Hydraulic Conductivity in a MAR Infiltration Pond | Ankita Prayag<br>*Lund University* |
| Geophysical Characterization of Groundwater Properties for Transport Infrastructure Development | Torleif Dahlin / Tina Martin<br>*Lund University* |

### Material and interfacial controls on IP

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| Combined Spectral Induced Polarization, Gain Phase and Vector Network Analyzer methods to investigate bentonite slurry | Thierry Bore<br>*The University of Queensland* |
| Characterization of corrosion effects of metallic spheres | Katrin Breede<br>*Clausthal University of Technology* |
| The effect of surface roughness on the spectral induced polarization response of metallic particles | Dennis Kreith<br>*Kiel University* |
| Hysteresis in complex conductivity of an opal-CT bearing diatomite under salinity diffusion | Ermis Proestakis<br>*Technical University of Denmark* |

### IP modelling and upscaling

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| Inductive or Galvanic IP? The pros and cons | James Macnae<br>*RMIT University and CD3D* |
| SIP modelling using effective medium theory and a GPU-accelerated finite volume Poisson–Nernst–Planck solver | Jean-Luc Gagnon<br>*Polytechnique Montréal* |
| Influence of particle-particle interaction on the induced-polarization response | Dennis Kreith<br>*Kiel University* |
| A model for the electrical polarization of microbial cells | Felix Keiser<br>*TU Braunschweig* |

## Poster presentations

Posters are up at all times in the adjacent Arctic Room.

| | Title | First author and affiliation |
|---|---|---|
| 1. | Adsorbed water as a surface-area indicator in induced polarization of smectite-rich biosiliceous mudstone | Shohreh Iraji<br>*Technical University of Denmark* |
| 2. | Spectral induced polarization signatures of zero valent iron as a function of particle size and corrosion | Zoe Vincent<br>*Pacific Northwest National Laboratory* |
| 3. | SIP-responses and effects of anisotropy and flint in bryozoan limestones | Sara Johansson<br>*Lund University* |
| 4. | Round-robin test of SIP laboratory measurements using electrical test networks | Tina Martin<br>*Lund University* |
| 5. | Review of broadband electrical behavior of clays: critical synthesis and comparison of experimental signatures and models | Louise Prétou<br>*BRGM / Université de Poitiers* |
| 6 | Evaluating the Effect of Surface Conduction on Archie's Saturation Exponent from Spectral Induced Polarization | Himadri Shekhar<br>*Rutgers University* |
| 7. | Surface Conductivity Dynamics During Electrokinetic Remediation | M. Vásconez-Maza<br>*British Geological Survey* |
| 8. | Numerical Simulations of the Influence of Soil properties on the shape factor | Felix Keiser<br>*Technical University Braunschweig* |
| 9. | A mesh-free machine learning approach for modelling microscale spectral induced polarization | Emilienne Rachel Kenko<br>*Polytechnique Montréal* |
| 10. | Contribution of the Induced Polarization method for hillslope characterization of deeply weathered crystalline terrains in Southeast Brazil | Marianna Tagnin<br>*University of São Paulo* |
| 11. | A multi-method geophysical approach for imaging groundwater flow in dikes: combining electrical resistivity, time-domain induced polarization, seismic refraction and self-potential | Jessy Richard<br>*Université Savoie Mont-Blanc / NAGA Geophysics* |
| 12. | The ice content of palsas and peat plateaus in Finnmark estimated with High-Frequency Induced Polarization | Raphael Schulz<br>*Technical University Braunschweig* |
| 13. | Anisotropic complex conductivity of the Manicouagan Reservoir lithologies: Preliminary results | Léopold Poussin<br>*Polytechnique Montréal* |

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| 14. | Spectral Induced Polarization Signatures of Rock samples from the Juomasuo Au-Co Deposit, Finland: Implications for Mineral Exploration | Bitnarae Kim<br>*BRGM / Geological Survey of Finland* |
| 15. | Concept of a SIP field measuring instrument with multiple decentralised transmitters | Tino Radic<br>*Radic Research* |

## Sponsors

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